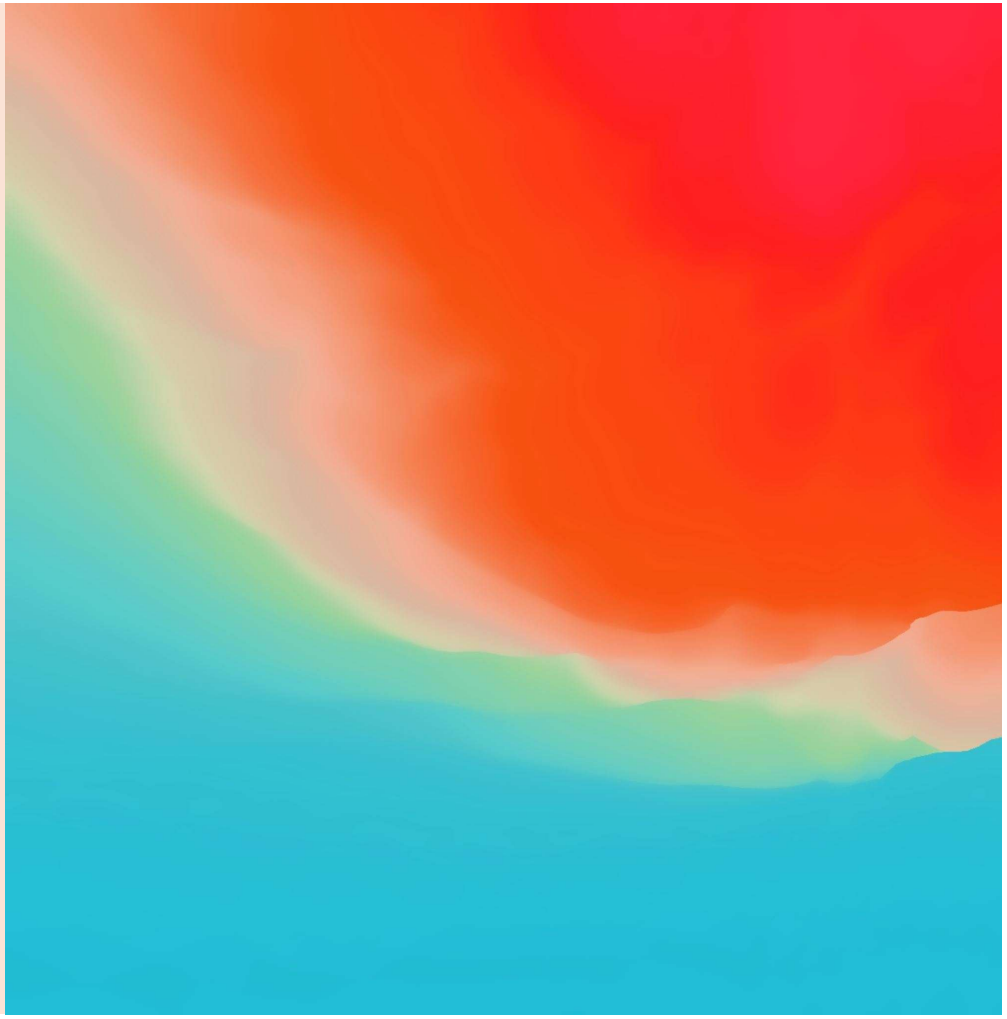
F18CE Week 4 Optimisation in multivariable functions



1. Global (absolute) vs local minimum/maximum.

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function. Suppose $(a, b) \in \mathbb{R}^2$. Then

(a, b) is a	Global		Local	
	Minimum	Maximum	Minimum	Maximum
if $f(a, b)$	$\leq f(x,\ y)$	$\geq f(x,\ y)$	$\leq f(x,\ y)$	$\geq f(x,\ y)$
	$\forall (x, y) \in \mathbb{R}^2$.		$\forall (x, y)$ in a neighbourhood of (a, b) .	

Remark: A global extremum is a local extremum, but a local extremum may not be a global extremum.

2. Rules of local extremum for function $f : \mathbb{R} \to \mathbb{R}$. Suppose n is odd such that $f^{(n)}(x = a) = 0$ and $f^{(n+1)}(x = a) \neq 0$. Then

(a) $(a, f(a))$ is a local minimum if $f^{(n+1)}(x = a) > 0$.

(b) $(a, f(a))$ is a local maximum if $f^{(n+1)}(x = a) < 0$.

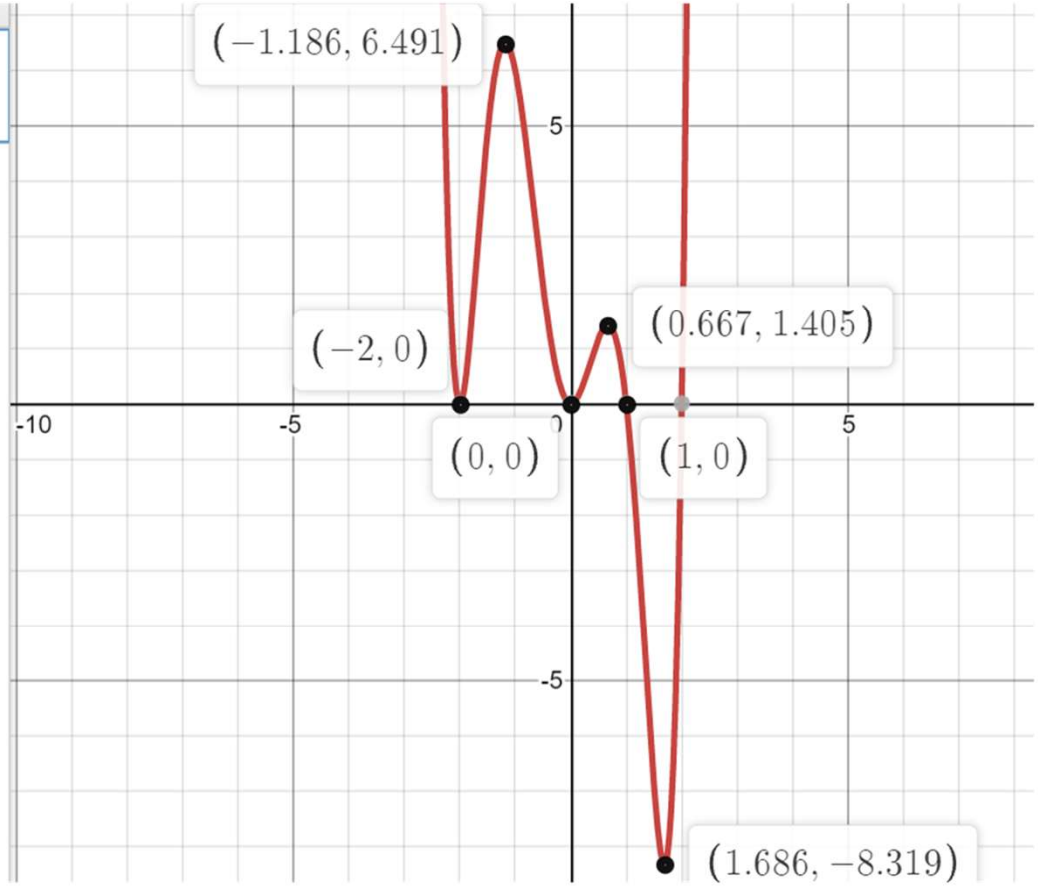
For example, $f(x) = x^4 \Rightarrow f^{(3)}(x) = 24x \Rightarrow f^{(3)}(0) = 0$ and $f^{(4)}(x) = 24 \Rightarrow f^{(4)}(0) = 24 > 0$. So $f(x) = x^4$ has a local minimum at (0, 0).

3. Rules of inflection point for function $f : \mathbb{R} \to \mathbb{R}$. Suppose n is even such that $f^{(n)}(x = a) = 0$ and $f^{(n+1)}(x = a) \neq 0$. Then $(a, f(a))$ is an inflection point.

For example, $f(x) = x^3 \Rightarrow f^{(2)}(x) = 6x \Rightarrow f^{(2)}(0) = 0$ and $f^{(3)}(x) = 6 \Rightarrow f^{(3)}(0) = 6 \neq 0$. So $f(x) = x^3$ has an inflection point at (0, 0).

4. Example. Determine the local and absolute extremum.

$y = (x-1)x^2(x+2)^2(x-2)$




Optimisation of
unconstrained
two-variable functions

Disclaimer

Let f be a multivariable function. Throughout the course, we shall assume that f is a class-C^1 function, unless otherwise stated.

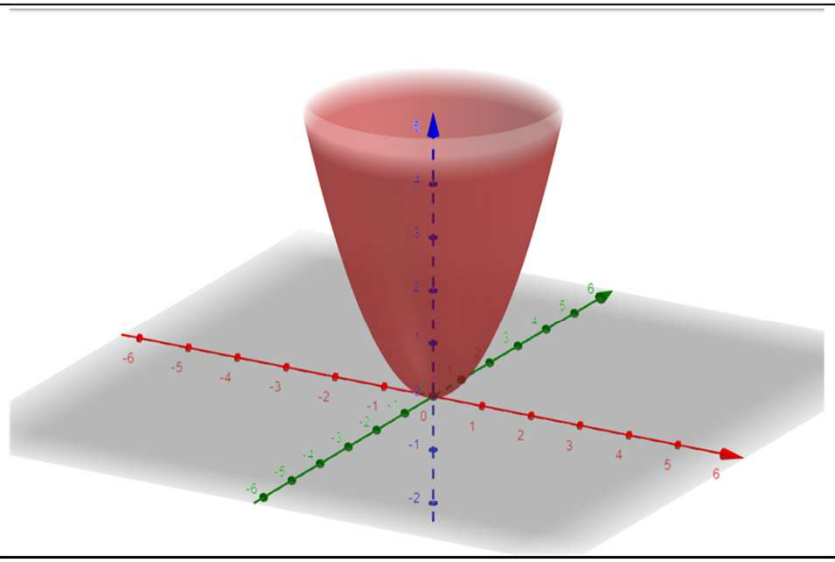
5. Stationary point. (Definition 5.2)

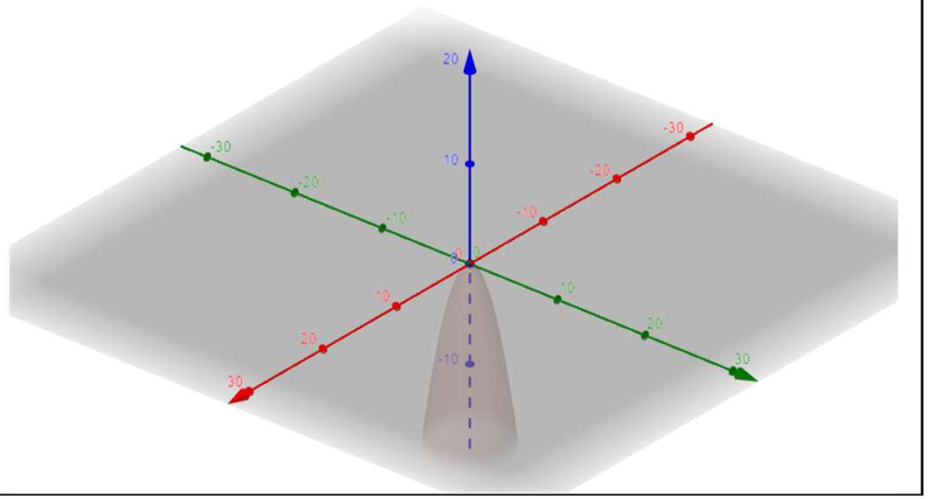
Let $f : \mathbb{R}^2 \to \mathbb{R}$. If $f_x(a,b) = f_y(a,b) = 0$, then $(a,\ b)$ is called a stationary point.

A stationary point in this case can be either a local minimum, local maximum or a saddle point.

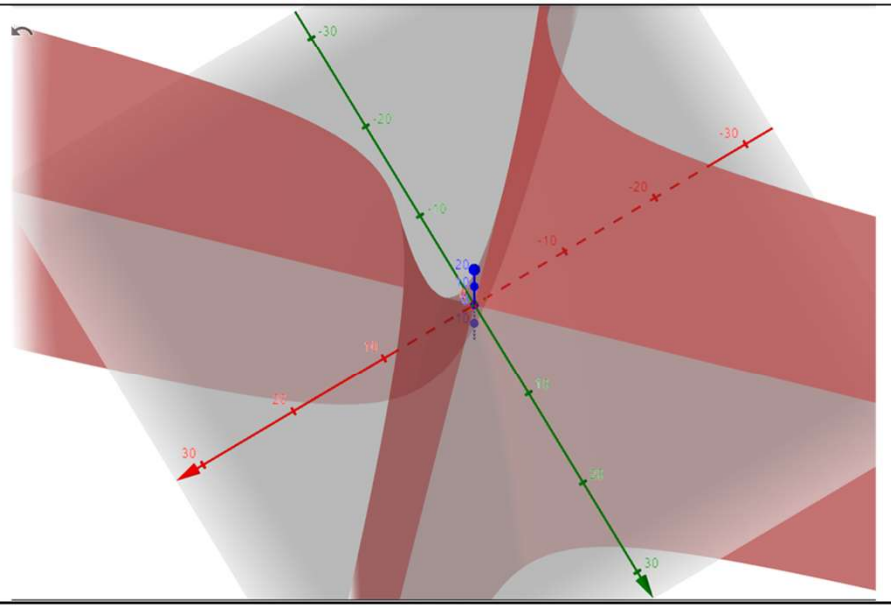
Remark: Stationary points also known as critical points

Example. (Example 5.3)

(a) For the function $f(x,y) = x^2 + y^2$, then (0,0) is a stationary point which happens to be a local minimum.	

(b) For the function $f(x,y) = -x^2 - y^2$, the (0,0) is a stationary point and is a local maximum.	

(c) For $f(x,y) = x^2 - y^2$, the point (0,0) is a stationary point and a saddle point.

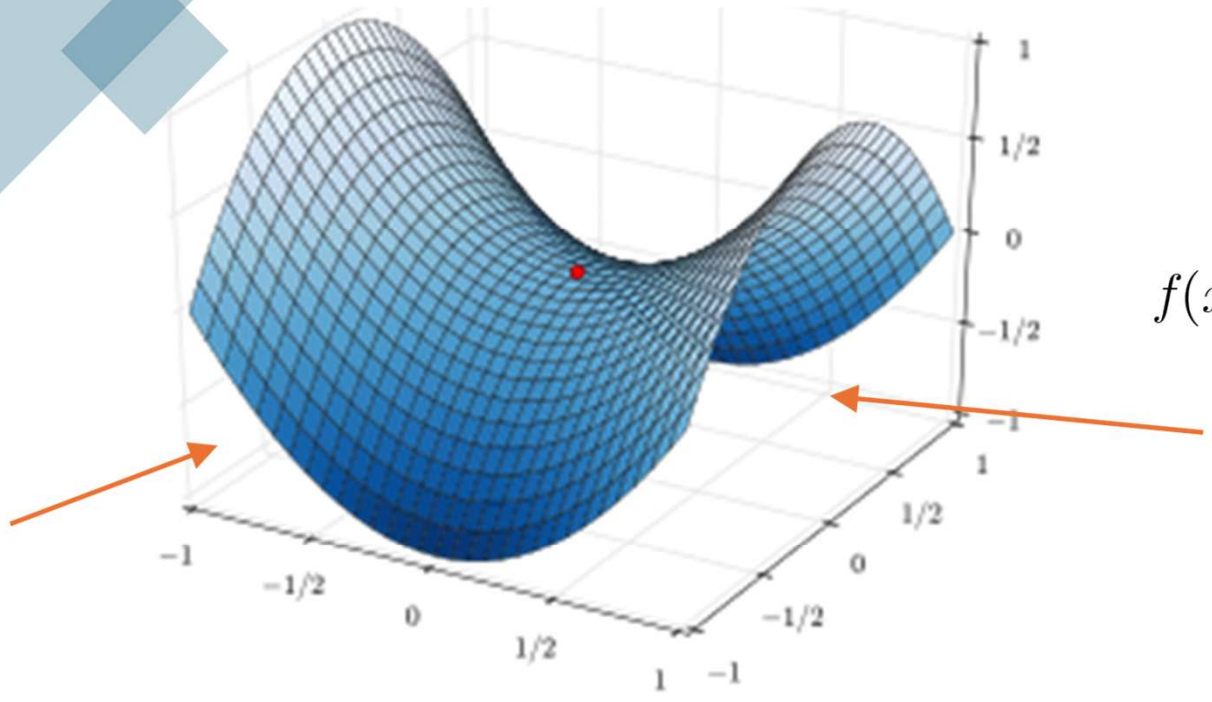


What is a saddle point?

Let $f(x,y) : \mathbb{R}^2 \to \mathbb{R}$ be a function such that $f \in C^1$.

A saddle point or *minimax* point (a,b) is a point on the surface of the graph of f that satisfies $\boxed{f_x(a,b) = 0 = f_y(a,b) \text{ and } D(a,b) < 0 \text{, where } D = f_{xx}f_{yy} - f_{xy}f_{yx}}$.

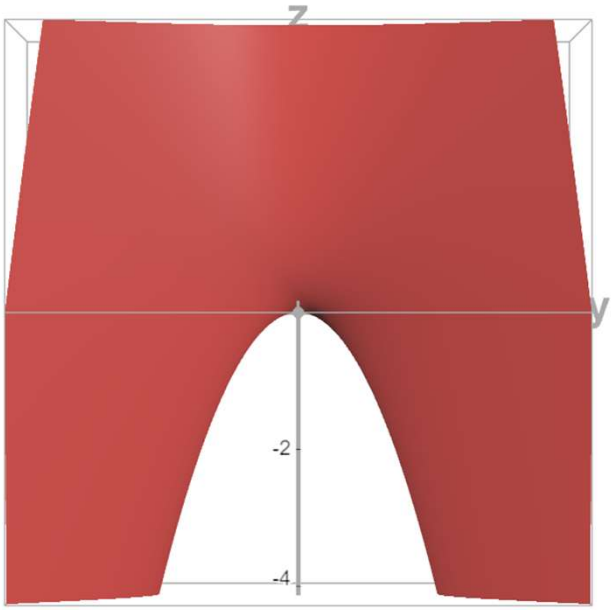
Geometrically, we see that the slope of the point are zeros when we view the graph of f in orthogonal directions (z vs x and z vs y).

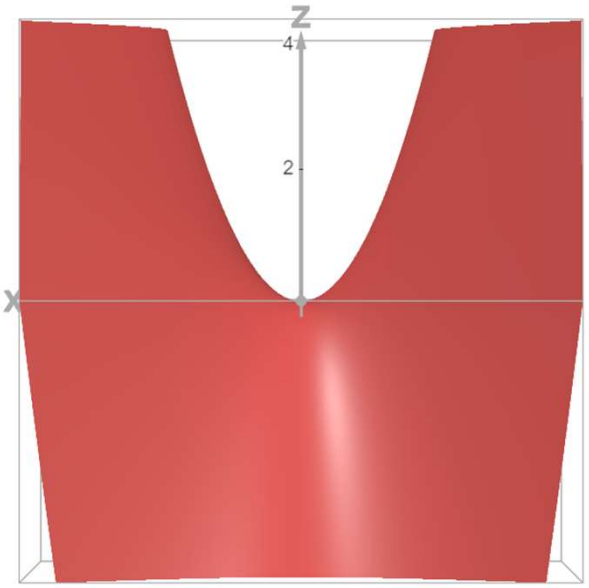


$$f(x, y) = x^2 - y^2$$

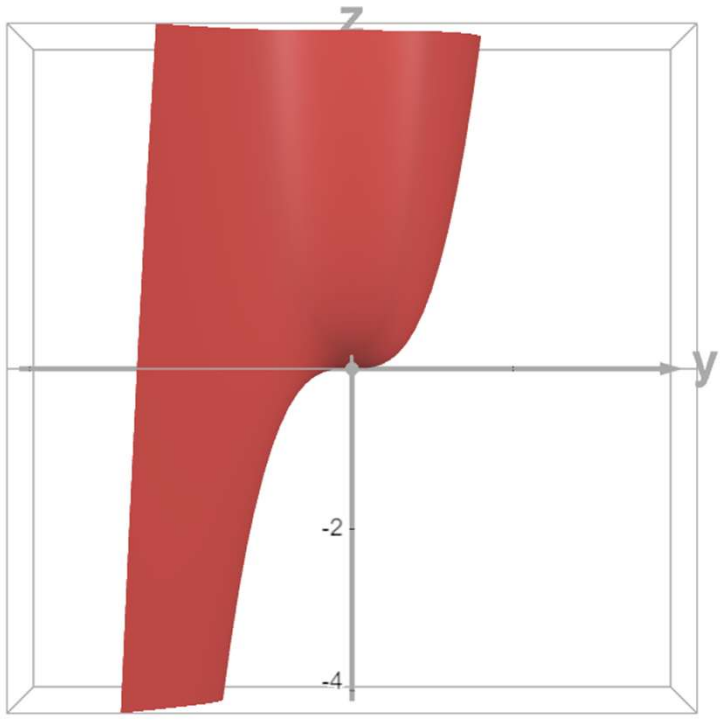
An example of a saddle point is the point looks like a local minimum (we call relative minimum) from one direction but local maximum (relative maximum) from the other direction.

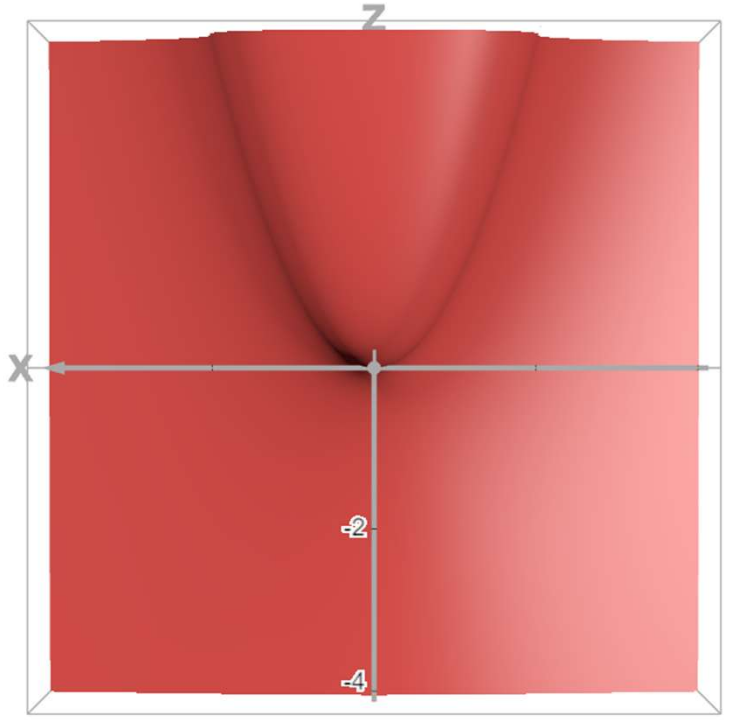
$$f(x,y) = x^2 - y^2$$





However, in general a saddle point need not be in this form. For example, the function $f(x, y) = x^2 + y^3$ has a saddle point at $(0, 0)$, but it does not have a relative maximum or relative minimum in the z vs y direction.





6. Second derivative test for two variables (D-test).

Let $D(x, y) = f_{xx}(x, y)f_{yy}(x, y) - f_{xy}(x, y)f_{yx}(x, y)$. Consider a stationary point (a, b):

Case 1. When $D(a, b) > 0$.

a) If $f_{xx}(a, b)$ (or $f_{yy}(a, b)$) < 0, then (a, b) is a local maximum.

b) If $f_{xx}(a, b)$ (or $f_{yy}(a, b)$) > 0, then (a, b) is a local minimum.

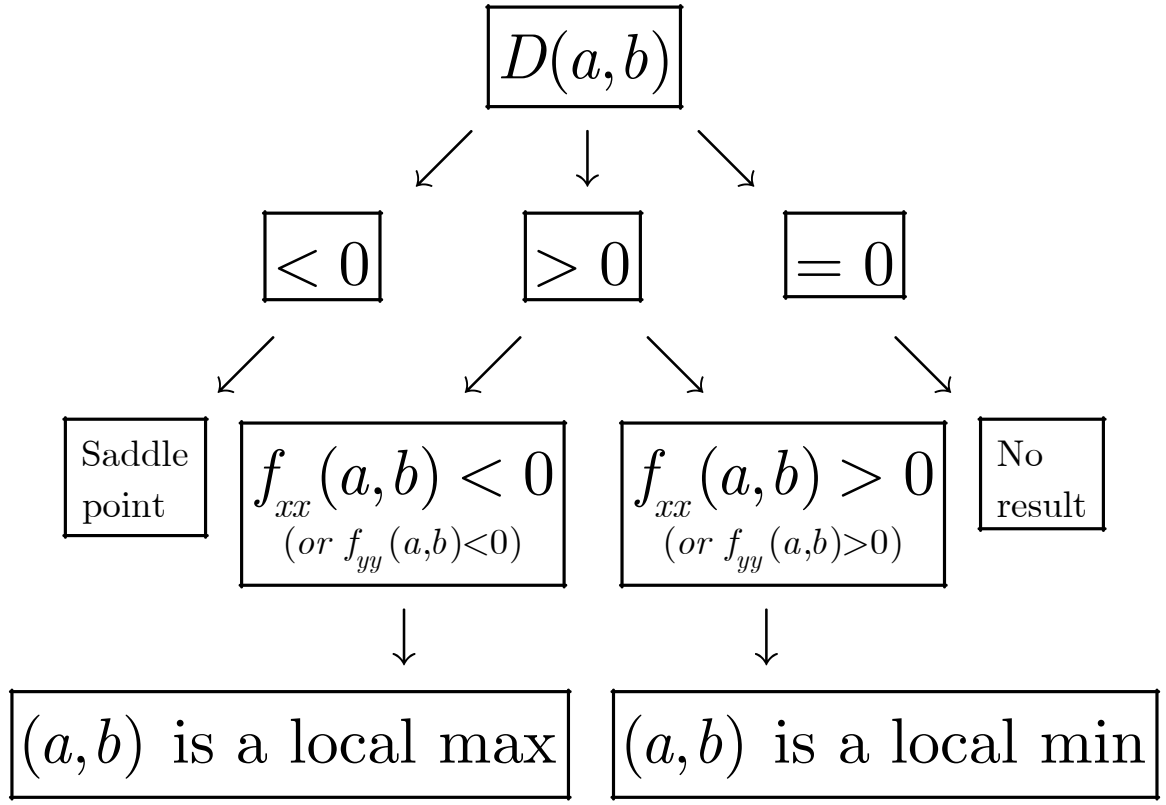
Case 2. If $D(a, b) < 0$, then (a, b) is a saddle point.

Case 3. If $D(a, b) = 0$, then the test is inconclusive.

$$\boxed{D = \det\begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}}$$

Optimisation of unconstrained function with two variables

Summary: If $f_{xy} = f_{yx}$ (when we can apply the mixed derivatives theorem).



7. Example 5.4: Find the stationary points and determine their nature.

(a) $f(x,y) = 2x^2 - xy + y^2 + 7x$

Solution. Short-cut: $f(x,y) = 2x^2 - xy + y^2 + 7x = (\frac{1}{2}x - y)^2 + \frac{7}{4}(x+2)^2 - 7 \geq -7$

we see that (–2, –1) is the local (also absolute) minimum where $f_{\min} = -7$.

$$f_x = 4x - y + 7 \qquad f_y = -x + 2y$$
$$f_{xx} = 4 \qquad f_{yy} = 2 \qquad f_{xy} = f_{yx} = -1$$

We find the stationary point.

$$\left.\begin{aligned} f_x &= 4x - y + 7 = 0 \\ f_y &= -x + 2y = 0 \end{aligned}\right\} \Rightarrow x = -2,\ y = -1$$

Using the D-test we write $D = f_{xx}f_{yy} - f_{xy}^2 = 4(2) - (-1)^2 > 0$ and $f_{xx} = 4 > 0$

(–2, –1) is a local minimum.

(b) $g(x,y) = xy - x^3y - xy^3$.

Solution. $g_x = y - 3x^2y - y^3$ $\quad$ $g_y = x - 3xy^2 - x^3$

$g_{xx} = -6xy$ $\quad$ $g_{yy} = -6xy$ $\quad$ $g_{xy} = g_{yx} = 1 - 3y^2 - 3x^2$

We find the stationary points.

$$g_x = y - 3x^2y - y^3 = y(1 - 3x^2 - y^2) = 0 \Rightarrow y = 0 \text{ or } 3x^2 + y^2 = 1$$

$$g_y = x - 3xy^2 - x^3 = x(1 - 3y^2 - x^2) = 0 \Rightarrow x = 0 \text{ or } 3y^2 + x^2 = 1$$

$$D = 36x^2y^2 - (1 - 3x^2 - 3y^2)^2 .$$

Case 1. $y = 0$ and $x = 0$. The point $(0, 0)$ is a saddle point because $D(0,0) = -1 < 0$.

Case 2. $y = 0$ and $3y^2 + x^2 = 1$. This gives us two points: $(1, 0)$ and $(-1, 0)$ and both are saddle points since $D(\pm 1, 0) = -4 < 0$.

$$D = 36x^2y^2 - (1 - 3x^2 - 3y^2)^2 .$$

Case 3. $x = 0$ and $3x^2 + y^2 = 1$. This gives us two points: $(0, \pm 1)$ and both are saddle points since $D(0, \pm 1) = -4 < 0$.

$$D = 36x^2y^2 - (1 - 3x^2 - 3y^2)^2 .$$

Case 4. $3y^2 + x^2 = 1$ and $3x^2 + y^2 = 1$. This gives us

$$3y^2 + x^2 = 3x^2 + y^2 \Rightarrow y^2 = x^2 \Rightarrow x^2 - y^2 = 0 \Rightarrow (x - y)(x + y) = 0 \Rightarrow x = \pm y$$

Then, $3y^2 + x^2 = 1 \Rightarrow 4y^2 = 1 \Rightarrow y = \pm\frac{1}{2}$.

Therefore, $(x, y) = (\pm\frac{1}{2}, \pm\frac{1}{2})$ (four solutions).

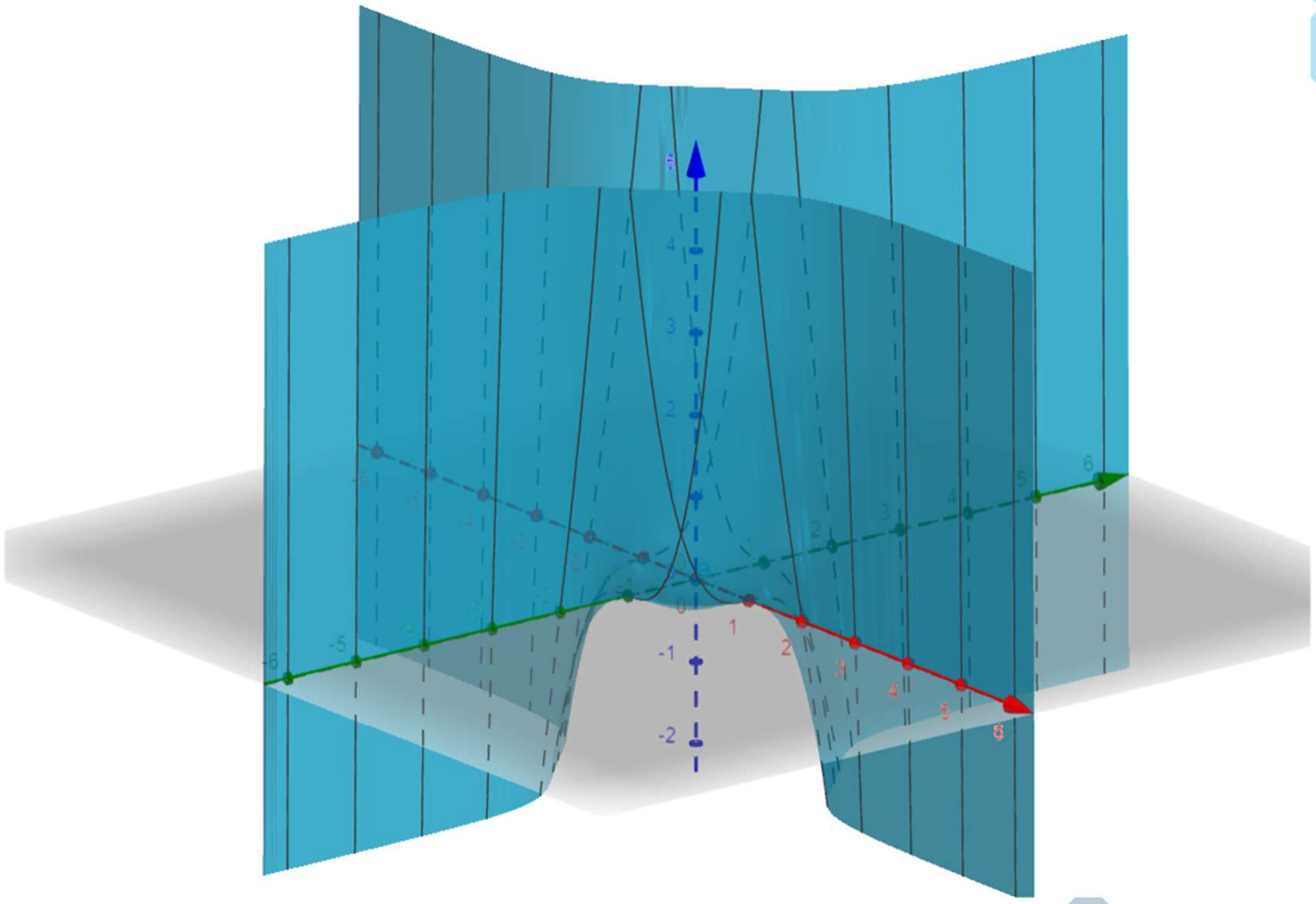
Points	$D = 36x^2y^2 - (1 - 3x^2 - 3y^2)^2$	g_{xx}	Nature of the points
(0.5, 0.5)	2	–1.5	Local max
(–0.5, 0.5)	2	1.5	Local min
(0.5, –0.5)	2	1.5	Local min
(–0.5, –0.5)	2	–1.5	Local max

Case 5. If $y = 0$ and $3x^2 + y^2 = 1$, then neither $x = 0$ nor $3y^2 + x^2 = 1$ can be satisfied.

Case 6. Similar to Case 5, if $x = 0$ and $3y^2 + x^2 = 1$, then neither $y = 0$ nor $3x^2 + y^2 = 1$ can be satisfied.

Case 7. In view of Case 5 and 6, it is impossible to have $x = 0$, $3y^2 + x^2 = 1$, $y = 0$ and $3x^2 + y^2 = 1$ to be satisfied at once.

Points	Nature of the points
$(0, 0), (0, \pm 1), (\pm 1, 0)$	Saddle points
$(0.5, 0.5), (-0.5, -0.5)$	Local max
$(-0.5, 0.5), (0.5, -0.5)$	Local min


z
4
3
2
-1
-2
-5
-4
1
2
3
4
5
6

(Example 5.5)

8. Consider $f(x,y) = (x-2)^2 - (y-1)^2 + 2(x+3y-5)^4$.

$$
\begin{aligned}
f_x &= 2(x-2) + 8(x+3y-5)^3 \\
f_y &= -2(y-1) + 24(x+3y-5)^3 \\
f_{xx} &= 2 + 24(x+3y-5)^2 \\
f_{yy} &= -2 + 216(x+3y-5)^2 \\
f_{xy} &= 72(x+3y-5)^2
\end{aligned}
$$

Let $u = x+3y-5$, then $D = (2+24u^2)(-2+216u^2) - (72u^2)^2$.

Now solving

$$2(x-2)+8u^3=0 \Rightarrow x-2+4u^3=0 \quad (1)$$

$$-2(y-1)+24u^3=0 \Rightarrow y-1-12u^3=0 \quad (2)$$

Take (1)×3 + (2): $3x+y=7$ (3)

Substitute (3) into (1): $x-2+4(x+21-9x-5)^3=0$

$$\Rightarrow x-2+4(-8x+16)^3=0$$

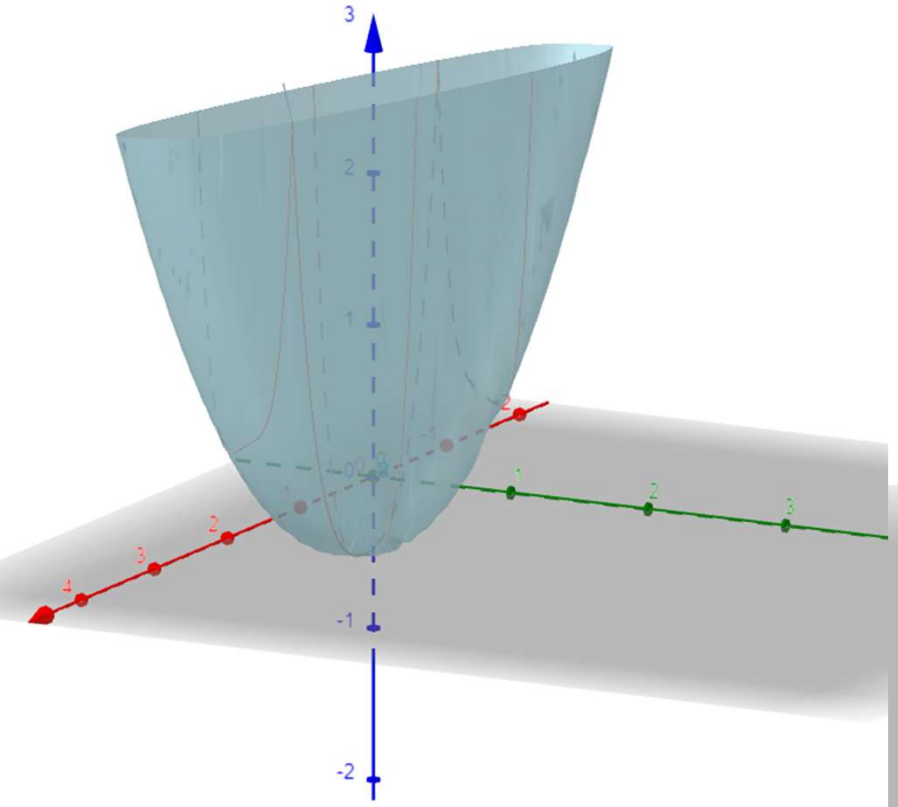
$$\Rightarrow (x-2)-4(8^3)(x-2)^3=0$$

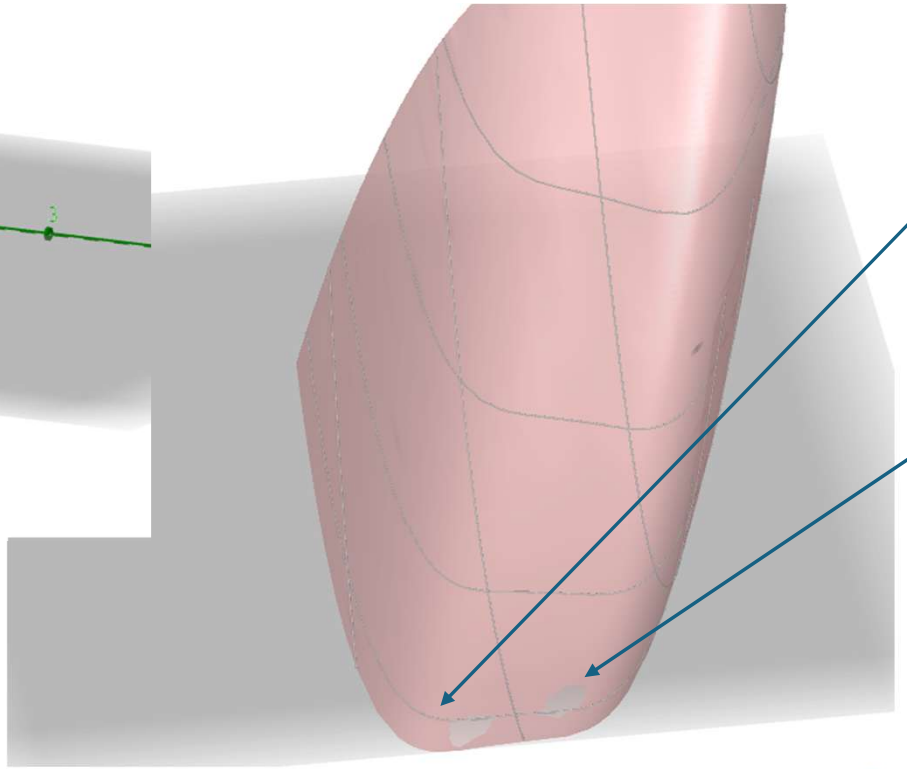
$$\Rightarrow (x-2)(1-2^{11}(x-2)^2)=0$$

$$\Rightarrow x=2,2\pm 2^{-11/2}$$

$$\therefore (x,y)=\left(2,1\right),\left(2+2^{-11/2},1-3(2^{-11/2})\right),\left(2-2^{-11/2},1+3(2^{-11/2})\right)$$

Points	u	D	f_{xx}	Nature of the points
(2, 1)	0	$-4 < 0$		Saddle
$(2+2^{-11/2}, 1-3(2^{-11/2}))$	–0.1767766953	$8 > 0$	> 0	Local min
$(2-2^{-11/2}, 1+3(2^{-11/2}))$	0.1767766953	$8 > 0$	> 0	Local min





The two local minimums

Question

Consider a function f (either single or multivariable).

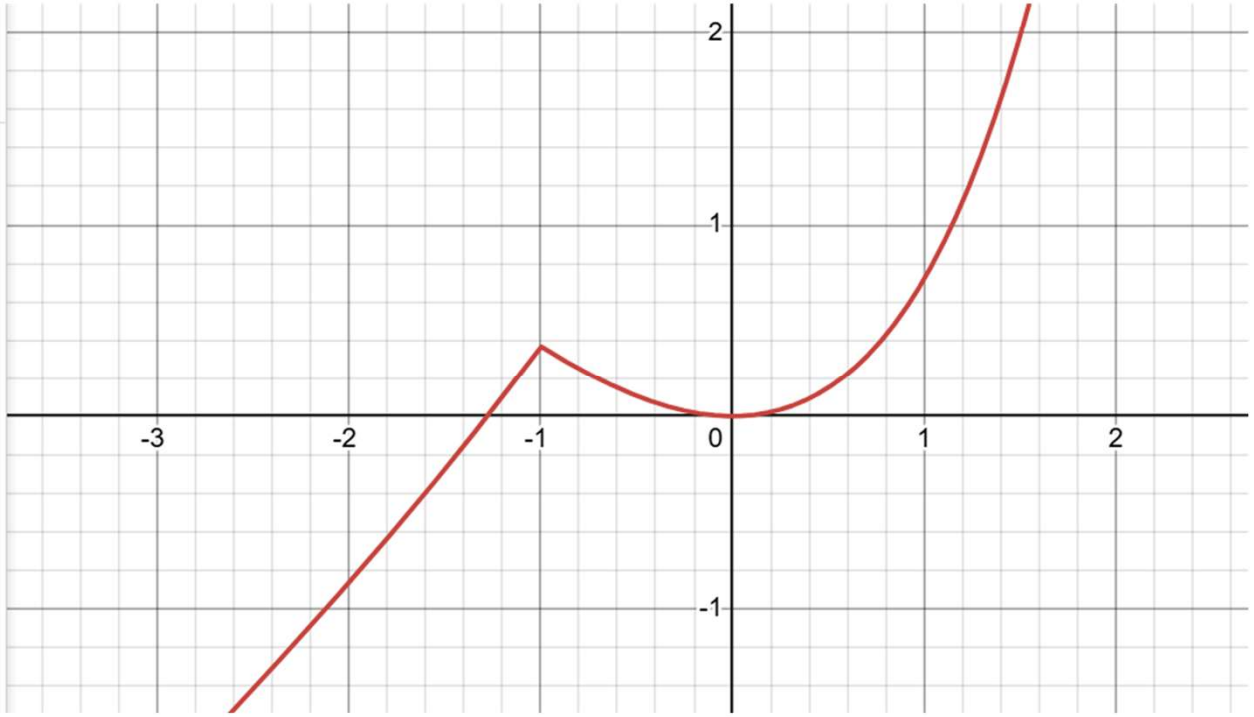
Suppose there exists only one local min (local max) for f, does it follow that the point is the absolute min (absolute max)?

Case 1. Single-variable function

If $f : \mathbb{R} \to \mathbb{R}$ is an everywhere differentiable function. Then the existence of unique critical point guarantees the absolute value.

However, if f is continuous but not everywhere differentiable, a counter example (when f is just continuous): $f(x) = e^x - \left|x + 1\right|$.

$e^x - |x+1|$



Remark (for single variable function)

When f is not everywhere differentiable. Let P be a point on f.

- If P is a stationary point and P is not an inflection point, then P must be a local min or local max point.
- However, a local min or local max point may not be a stationary point.

Case 2. Two-variable function

Let $f(x,y) = e^{3x} + y^3 - 3ye^x$. This is differentiable for all $(x,\ y)$.

In looking for critical points, we solve

$$f_x = 3e^{3x} - 3ye^x = 0 \Rightarrow e^x\left(e^{2x} - y\right) = 0 \Rightarrow y = e^{2x} > 0\ *$$
$$f_y = 3y^2 - 3e^x = 0 \Rightarrow y^2 = e^x$$

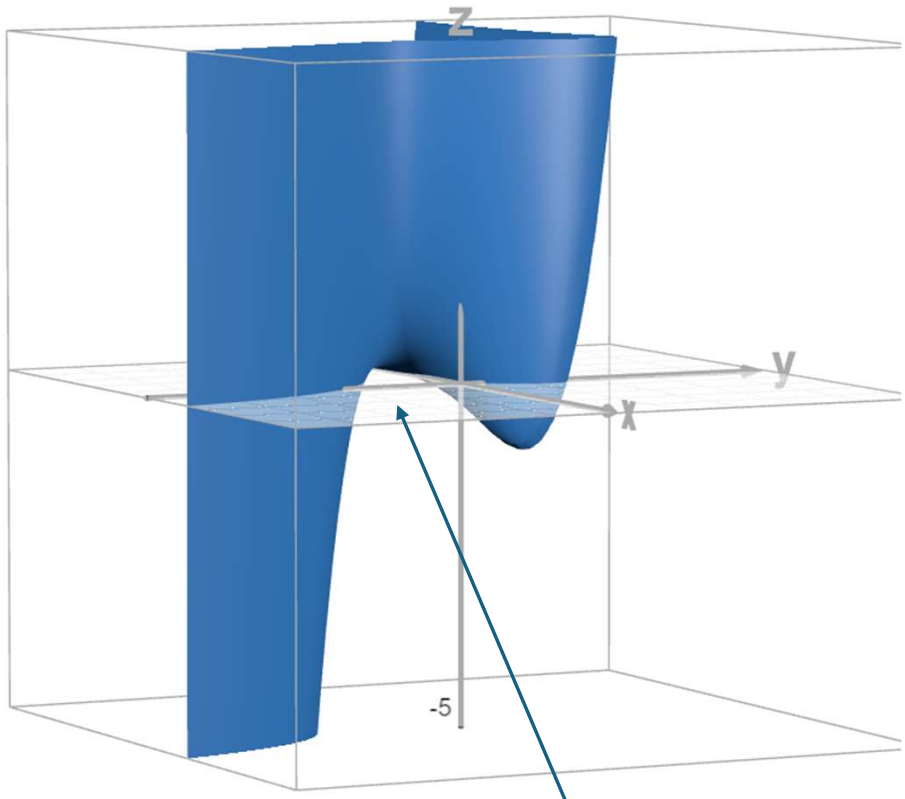
This implies that $y^2 = e^{4x} = e^x \Rightarrow x = 0$, and so $y = 1$ (as $y > 0$ from *).

The second derivative test shows that this point is a local minimum for f.

$$f_{xx} = 9e^{3x} - 3ye^{x} \Rightarrow f_{xx}(0,1) = 6 > 0$$
$$f_{yy} = 6y \Rightarrow f_{yy}(0,1) = 6 \qquad \text{implies that } D = 6^2 - (-3)^2 > 0\,.$$
$$f_{xy} = -3e^{x} \Rightarrow f_{xy}(0,1) = -3$$

However, $f(0,-3) = -17 < f(0,1) = -1$, so it is not an absolute minimum.

$$z = e^{3x} + y^3 - 3ye^x$$



Interestingly this is neither a local max nor a saddle point